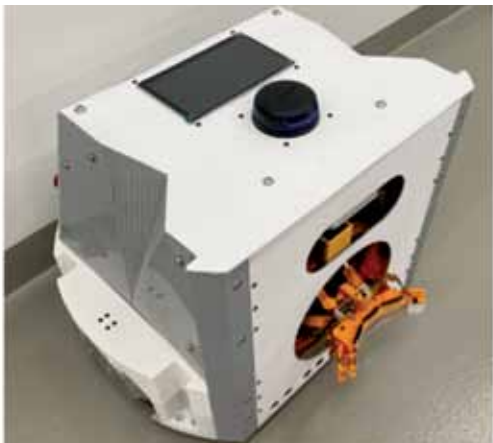


Charging system for autonomous robots

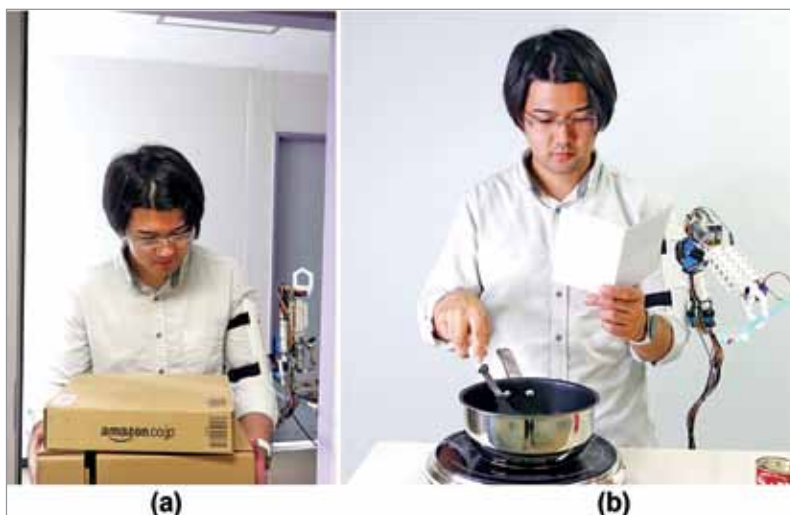


Autonomous robots are expected to be a reality in the near future. However, one of the major challenges for realising swarms of autonomous robots is if any of them depletes its power, it can pose a serious obstacle. Therefore, researchers from Skolkovo Institute of Science and Technology in Russia have developed a system called MobileCharger, an autonomous robotic system designed to charge other robots as they complete their missions. The idea is to reach faulty robots, or robots with a depleted battery power, and assist them. The system is similar to aerial refueling, in which airplanes are fuelled during flight operation so that they can remain in the air longer. The MobileCharger incorporates a system that can position electrodes to sustain charging in situations where a target robot is not properly aligned to the charger.

The prototype of MobileCharger (Credit: <https://techxplore.com>)

Compact robotic limb for human augmentation

There has always been an interest in human augmentation technologies to enhance human capabilities with information and robotics approaches. Previously-developed robotic limbs have been quite heavy and acquired a lot of space. Now, researchers from the Japan Advanced Institute of Science and Technology and University of Tokyo have developed a compact robotic limb called AugLimb that could support humans during a variety of everyday tasks. The compact robotic limb can fold into a small volume without interrupting the user's daily activities, especially for long-time usage. The AugLimb can stretch approximately 2.5 times longer than a human's average forearm length. It has seven degrees of freedom and can be easily fabricated and implemented using 3D printers and motors.



A compact robotic limb to support humans during everyday activities (Credit: www.jaist.ac.jp)



3D-printed personal wireless wearables

Wearable sensors that help monitor everything from step count to heart rate are nearly ubiquitous. But for scenarios such as measuring the onset of frailty in older adults, promptly diagnosing deadly diseases, testing the efficacy of new drugs, or tracking the performance of professional athletes, medical-grade devices are needed. University of Arizona engineers have now developed a type of wearable they call a bio-symbiotic device, which is not only custom 3D-printed and based on body scans of wearers but can operate continuously using a combination of wireless power transfer and compact energy storage. And because these bio-symbiotic devices are custom fitted to the wearer, they are also highly sensitive.

University of Arizona engineers have developed a way to 3D-print medical-grade wearable devices, based on body scans of the wearer (Credit: <https://news.arizona.edu>)